

Proposal Outline: DUE March 27th at Midnight

(LastName_Assignment6.doc on D2L)

1) Revise Title

2) Revise abstract.

Adhere to word limit: The abstract must be written in standard ASCII and should be no longer than 20 lines of 85 characters of text. Use the HST/JWST Templates.

Science Justification will need to be 3 pages in total. This includes 1 figure, but not the bibliography.

Science Justification Outline

3) Provide an outline for your proposal (BULLETED POINTS that answer each of the below)

Section 1: Facts, Problem & Summary paragraph (3/4 of a page)

- a) What is the descriptive heading? It must be a *statement*
- b) 1 fact-based paragraph:
 - i) State the fact leading the paragraph (the conclusion) – this should follow from the abstract.
 - ii) What would be supporting facts (1-3)? Provide at least one paper citation per supporting fact.
 - iii) What is the nuanced conclusion for the paragraph? This is the importance of the fact that should set the stage for the problem (should follow from abstract)
- c) Problem paragraph:
 - iv) What is “the problem” statement?
 - v) What references are needed to support the problem?
 - vi) Why is the problem critical to answer now? (nuanced conclusion)
- d) Synopsis paragraph:
 - vii) What is the proposal? (identify key component) “We propose”
 - viii) Why the facility? This can be related to the funding call.
 - ix) What is the broader impact ? Community deliverables?

Section 2: The Problem & Motivation/Narrative of the proposal

- a) Identify a descriptive heading that states the problem
- b) What fact do you need to start with to set up the problem?
- c) What references support the problem?
- d) State why solving this problem advances the subfield.
- e) Identify the “key component” missing to solve the problem.
- f) State that this program will generate the “key component” to solve the problem.

Section 3: The Strategy: HST/JWST + Target Solves the Problem (Generate the “Key Component”)

- g) Create a descriptive heading that states the strategy → “This Proposal: Observe/Model This Thing”
- a) What are you targeting ? (imaging/spectra/archival data of **what astronomical object**, or **what physical mechanism/ simulations/data set**)
- b) Why this facility? (Why HST/JWST?)
- c) Outline (bullets) the strategy to generate the key component
 - For GO: describe the proposed observations *briefly/succinctly*
 - For AR/Theory: describe *briefly* the analysis method/proposed usage of simulations/codes and relation to existing/future HST/JWST data sets.
 - You must explain how the figure will be used to support the strategy
- d) Outline the products of the program (tangible outcomes)
- e) How do these products solve the problem?

Include a Figure that explains the strategy/target/solution

Add here your (revised) Figure from Assignment 4 (or you must create a new one if your proposal is different)

Figure and caption instructions are in Assignment 4.

Section 4: Importance and Broader Implications

- a) Create a heading that states the broader impact/importance of the solution.
- b) What is the **high level** importance of the proposed products of your study to your **subfield** ?
How will your study advance the subfield? (this is a like a summary statement)
- c) Repeat Suitability:
 - i) Why now?
 - ii) How does this program relate to the funding call?
Which **key science goals** of HST/JWST does your proposal connect to ?
 - JWST: <https://webb.nasa.gov/content/science/index.html>
 - HST: <https://www.stsci.edu/hst/about/key-science-themes>
[if applicable -> Observational] Repeat Why this facility?
- d) How does this study increase the **legacy value** of the facility/agency?
 - What **other subfields** are advanced by these observations (GO) or data analysis/code/simulations (AR) ? [Broader Impact]And/OR
 - Do the products of your proposal **serve the community** at large?